

Notes & Summary

$$\mathbf{r} = (x, y, z) \text{ or } (r, \theta, \phi) \text{ or } (r, \theta, z) \quad \mathbf{v} = \frac{d\mathbf{r}}{dt} \text{ velocity}$$

$$\mathbf{q} = (q_1, \dots, q_s)^T \text{ Generalized coordinates} \quad \dot{\mathbf{q}} = \frac{d\mathbf{q}}{dt}$$

$$\text{Action } S = \int_{t_0}^{t_1} L(\mathbf{q}, \dot{\mathbf{q}}, t) dt, \quad L = \overset{\text{Kinetic Energy}}{T} - \overset{\text{Potential Energy}}{U}$$

$$\delta S = 0 \quad \left\{ \begin{array}{l} S_{\mathbf{q}(t_0)} = S_{\mathbf{q}(t_1)} = 0 \end{array} \right\} \Rightarrow \boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = 0} \quad (1) \quad \frac{\partial L}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial L}{\partial q_1} \\ \vdots \\ \frac{\partial L}{\partial q_s} \end{bmatrix}$$

Euler-Lagrange Eq

* - System of N particles: $T = \frac{1}{2} \sum_{\alpha} m_{\alpha} v_{\alpha}^2, U = U(\mathbf{r}_1, \dots, \mathbf{r}_N)$

$$v_{\alpha} = \frac{||d\mathbf{r}||}{dt}, \quad d\mathbf{r}^2 = \begin{cases} dx^2 + dy^2 + dz^2 & \text{Cartesian} \\ dr^2 + r^2 d\theta^2 + dz^2 & \text{Cylindrical} \\ dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 & \text{Spherical} \end{cases}$$

$$(1) \Rightarrow m_{\alpha} \frac{d\mathbf{v}_{\alpha}}{dt} = - \frac{\partial U}{\partial \mathbf{r}_{\alpha}} \quad (\text{Newton}) \quad \vec{\mathbf{F}} = - \begin{bmatrix} \frac{\partial U}{\partial r_1} \\ \vdots \\ \frac{\partial U}{\partial r_N} \end{bmatrix}$$

→ Galileo principle of Relativity:

a particle moving freely in an inertial frame of reference $L = L(\dot{\mathbf{q}}) = L(||\mathbf{v}||^2) = \sum_n a_n ||\mathbf{v}||^{2n}$

$$\frac{\partial L}{\partial \mathbf{q}} = 0 = \frac{\partial L}{\partial \mathbf{r}} \quad \& \quad \frac{\partial L}{\partial \mathbf{v}} = \sum_n b_n ||\mathbf{v}||^{2n-1}$$

$$E.L.E \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = 0 \Rightarrow \boxed{\mathbf{v} = \text{cte}}$$

Law of inertia: a particle moving freely in its inertial frame of reference has constant velocity

Galileo transformation: $\mathcal{F} \rightarrow \mathcal{F}'$

$$\mathbf{r}' = \mathbf{r} + \mathbf{v}t \quad \& \quad t = t' \quad (\text{time is absolute \& same in both frames})$$

$\mathcal{F}'$ moves with respect to $\mathcal{F}$ with velocity $\mathbf{v}$

$$\begin{aligned} L(||\mathbf{v}'||^2) &= L(||\mathbf{v} + \boldsymbol{\varepsilon}||^2) = L(||\mathbf{v}||^2 + 2\langle \mathbf{v}, \boldsymbol{\varepsilon} \rangle + ||\boldsymbol{\varepsilon}||^2) \\ &= L(||\mathbf{v}||^2) + 2 \frac{\partial L}{\partial ||\mathbf{v}||^2} \langle \mathbf{v}, \boldsymbol{\varepsilon} \rangle + \text{neglected higher order terms} \end{aligned}$$

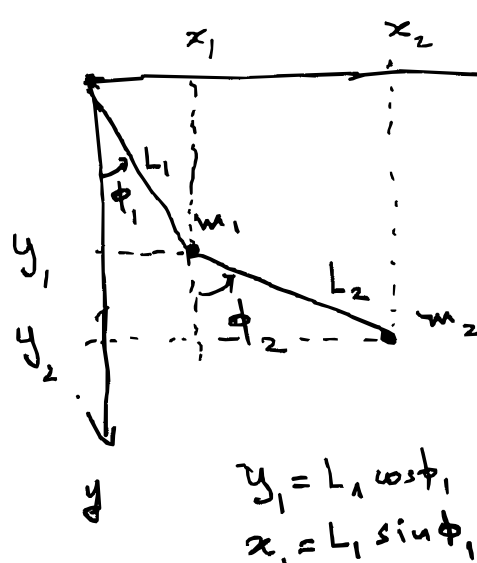
Galileo Principle of Relativity: Experiment shows that the frames $\mathcal{F}$ & $\mathcal{F}'$ are equivalent the laws of motion are the same in both frames

$$\frac{\partial L'}{\partial \mathbf{v}} = \frac{\partial L}{\partial \mathbf{v}} + 2 \frac{\partial L}{\partial ||\mathbf{v}||^2} \boldsymbol{\varepsilon} \quad \left(\begin{array}{l} A \& B \\ L_{A+B} = L_A + L_B \\ \text{if no interaction} \end{array} \right)$$

$$\frac{d}{dt} \left(\frac{\partial L'}{\partial \mathbf{v}} \right) = \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) \Rightarrow 2 \frac{d}{dt} \left(\frac{\partial L}{\partial ||\mathbf{v}||^2} \right) = 0$$

$$\Rightarrow \boxed{L(||\mathbf{v}||^2) = \frac{1}{2} m ||\mathbf{v}||^2}$$

Problem 1: Double Pendulum



$$y_1 = L_1 \cos \phi_1 \quad y_2 = L_1 \cos \phi_1 + L_2 \cos \phi_2$$

$$x_1 = L_1 \sin \phi_1 \quad x_2 = L_1 \sin \phi_1 + L_2 \sin \phi_2$$

Potential Energies: "Gravitation"

$$-\vec{\nabla} U_1 = m_1 g \vec{e}_y \Rightarrow U_1(y_1) = -m_1 g y_1 + \text{cte}$$

$$-\vec{\nabla} U_2 = m_2 g \vec{e}_y \Rightarrow U_2(y_2) = -m_2 g y_2 + \text{cte}$$

$$U_1(\phi_1) = -m_1 g L_1 \cos \phi_1 + \text{cte}_1$$

$$U_2(\phi_1, \phi_2) = -m_2 g L_1 \cos \phi_1 - m_2 g L_2 \cos \phi_2 + \text{cte}_2$$

$$\text{Initial condition: } \phi_1 = \phi_2 = 0 \quad U_1 = U_2 = 0$$

$$\text{cte}_1 = m_1 g L_1 \quad \text{cte}_2 = m_2 g (L_1 + L_2)$$

Kinetic Energies:

$$T_1 = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2)$$

$$= \frac{1}{2} m_1 L_1^2 \dot{\phi}_1^2$$

$$T_2 = \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

$$\dot{x}_2 = L_1 \dot{\phi}_1 \cos \phi_1 + L_2 \dot{\phi}_2 \cos \phi_2$$

$$\dot{y}_2 = -L_1 \dot{\phi}_1 \sin \phi_1 - L_2 \dot{\phi}_2 \sin \phi_2$$

$$\dot{x}_2^2 + \dot{y}_2^2 = L_1^2 \dot{\phi}_1^2 + L_2^2 \dot{\phi}_2^2 + 2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)$$

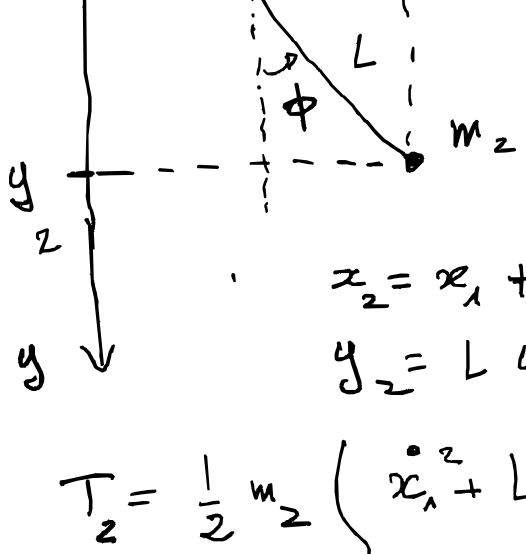
$$L(\phi_1, \phi_2) = \frac{1}{2} m_1 L_1^2 \dot{\phi}_1^2 + \frac{1}{2} m_2 L_1^2 \dot{\phi}_1^2 + \frac{1}{2} m_2 L_2^2 \dot{\phi}_2^2$$

$$+ m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)$$

$$- m_1 g L_1 (1 - \cos \phi_1)$$

$$- m_2 g L_2 (1 - \cos \phi_1) - m_2 g L_2 (1 - \cos \phi_2)$$

Problem 2:



* Particle of mass m_2 :

$$T_2 = \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

$$x_2 = x_1 + L \sin \phi$$

$$y_2 = L \cos \phi$$

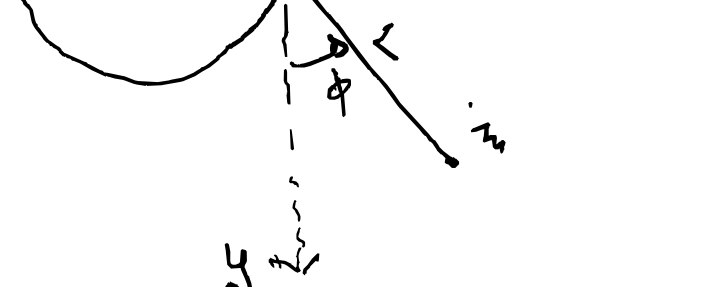
$$T_2 = \frac{1}{2} m_2 (\dot{x}_1^2 + L^2 \dot{\phi}^2 + 2 L \dot{x}_1 \dot{\phi} \cos \phi)$$

$$-\vec{\nabla} U_2 = m_2 g \vec{e}_y \Rightarrow U_2(y_2) = -m_2 g y_2 + \text{cte}$$

$$U_2(y_2) = m_2 g L (1 - \cos \phi) \quad (U_2(L) = 0)$$

$$L = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 (\dot{x}_1^2 + L^2 \dot{\phi}^2 + 2 L \dot{x}_1 \dot{\phi} \cos \phi) - m_2 g L (1 - \cos \phi)$$

Problem 3:



a). Uniform oscillations:

$$x = a \cos \gamma t + L \sin \phi$$

$$y = -a \sin \gamma t + L \cos \phi$$

$$\dot{x} = -a \gamma \sin \gamma t + L \dot{\phi} \cos \phi$$

$$\dot{y} = -a \gamma \cos \gamma t - L \dot{\phi} \sin \phi$$

$$\dot{x}^2 + \dot{y}^2 = a^2 \gamma^2 + L^2 \dot{\phi}^2 - 2 a \gamma L \dot{\phi} \sin \gamma t \cos \phi$$

$$+ 2 a \gamma L \dot{\phi} \cos \gamma t \sin \phi$$

$$= a^2 \gamma^2 + L^2 \dot{\phi}^2 + 2 a \gamma L \dot{\phi} \sin(\phi - \gamma t)$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$= \frac{1}{2} m a^2 \gamma^2 + \frac{1}{2} m L^2 \dot{\phi}^2 + m a L \gamma \dot{\phi} \sin(\phi - \gamma t)$$

$$L = \frac{1}{2} m L^2 \dot{\phi}^2 + m a L \gamma \dot{\phi} \sin(\phi - \gamma t)$$

$$- m g L (1 - \cos \phi)$$

b). horizontal:

$$x = a \cos \gamma t + L \sin \phi$$

$$y = L \cos \phi$$

$$\dot{x} = -a \gamma \sin \gamma t + L \dot{\phi} \cos \phi$$

$$\dot{y} = -L \dot{\phi} \sin \phi$$

$$\dot{x}^2 + \dot{y}^2 = a^2 \gamma^2 (\sin \gamma t)^2 + L^2 \dot{\phi}^2 - 2 L \dot{\phi} a \gamma \sin \gamma t$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

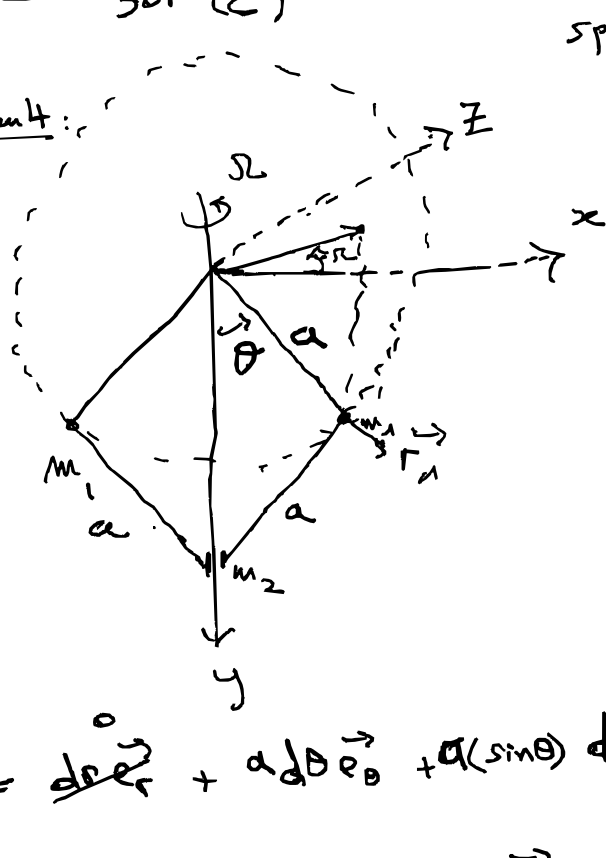
$$= \frac{1}{2} m a^2 \gamma^2 (\sin \gamma t)^2 + L^2 \dot{\phi}^2 - 2 L \dot{\phi} a \gamma \sin \gamma t \cos \phi$$

$$L = \frac{1}{2} m L^2 \dot{\phi}^2 - m L a \gamma \dot{\phi} \sin \gamma t \cos \phi + m g L \cos \phi$$

$$f(t) = a^2 \gamma^2 (\sin \gamma t)^2 \text{ is a total time derivative so it can be omitted from the Lagrangian}$$

same for (c)

Problem 4:



spherical coordinates

$$d\vec{s}_1 = d\vec{e}_r + a d\theta \vec{e}_\theta + a(\sin \theta) d\phi \vec{e}_\phi$$

$$\frac{d\vec{s}_1}{dt} = a \dot{\theta} \vec{e}_\theta + a \sin \theta \dot{\phi} \vec{e}_\phi \quad \left(\frac{d\phi}{dt} = \Omega \right)$$

angular velocity constant

$$T_1 = \frac{1}{2} m_1 \left| \frac{d\vec{s}_1}{dt} \right|^2$$

$$= \frac{1}{2} m_1 \{ a^2 \dot{\theta}^2 + a^2 (\sin \theta)^2 \Omega^2 \}$$

$$T_2 = \frac{1}{2} m_2 \dot{y}_2^2$$

$$y_2 = 2a \cos \theta \Rightarrow \dot{y}_2 = -2a \dot{\theta} \sin \theta$$

$$T_2 = \frac{1}{2} m_2 4 a^2 \dot{\theta}^2 (\sin \theta)^2 = 2 m_2 a^2 \dot{\theta}^2 (\sin \theta)^2$$

$$U_1 = -m_1 g a \cos \theta$$

$$U_2 = -m_2 g a \cos \theta$$

$$L = -q U_1 - U_2 + q T_1 + T_2$$

$$= L(m_1 + m_2) a \cos \theta + m_1 \{ a^2 \dot{\theta}^2 + a^2 (\sin \theta)^2 \Omega^2 \}$$

$$+ 2 m_2 a^2 \dot{\theta}^2 (\sin \theta)^2$$

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Summary & Notes: Chapter II Conservatⁿ Laws

total Derivative Lagrangian:

$$\frac{dL}{dt} = \sum_i \frac{\partial L}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} \quad (\text{E.L.E}) \quad \text{substituting}$$

$$\frac{dL}{dt} = \sum_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{d}{dt} \dot{q}_i$$

$$= \sum_i \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right)$$

$$\frac{d}{dt} \left(\underbrace{\sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i}}_E - L \right) = 0$$

E: Energy of the system

Conservation of Energy $\boxed{\frac{d}{dt} E(q, \dot{q}) = 0}$

in chapter 1: $L(q, \dot{q}) = T(q, \dot{q}) - U(q)$

$$\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial T}{\partial \dot{q}_i} \Rightarrow E = \sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} - T + U$$

T is a quadratic function:

$$T(q, \dot{q}) = \frac{1}{2} \sum_{ij} \alpha_{ij}(q) \dot{q}_i \dot{q}_j$$

$$\Rightarrow \sum_k \dot{q}_k \frac{\partial T}{\partial \dot{q}_k} = 2T$$

$$\boxed{E = T(q, \dot{q}) + U(q)}$$

Conclusions:

* for a closed system in Cartesian Coord

$$E = \frac{1}{2} \sum_a m_a v_a^2 + U(r_1, \dots, r_A)$$

* Additivity for two systems A & B

$$E_{A+B} = E_A + E_B$$

→ Conservation of Momentum: (translation invariance)

if we displace the whole system by a fixed amount the system remains unchanged!

$$r_a \rightarrow r_a + \varepsilon \quad \text{for each particle } a$$

$$\delta L = \sum_a \left\langle \frac{\partial L}{\partial r_a}, \delta r_a \right\rangle = \left\langle \sum_a \frac{\partial L}{\partial r_a}, \varepsilon \right\rangle$$

$$\delta L = 0 \Rightarrow \sum_a \frac{\partial L}{\partial r_a} = 0$$

$$\Rightarrow \sum_a \frac{d}{dt} \left(\frac{\partial L}{\partial v_a} \right) = 0$$

$$L = \frac{1}{2} \sum_a m_a v_a^2 - U(r_1, \dots, r_A)$$

$$\frac{\partial L}{\partial v_a} = m_a v_a = p_a : \text{momentum of particle } a$$

$$\Rightarrow \frac{d}{dt} \left(\underbrace{\sum_a p_a}_P \right) = 0$$

P: momentum of the system

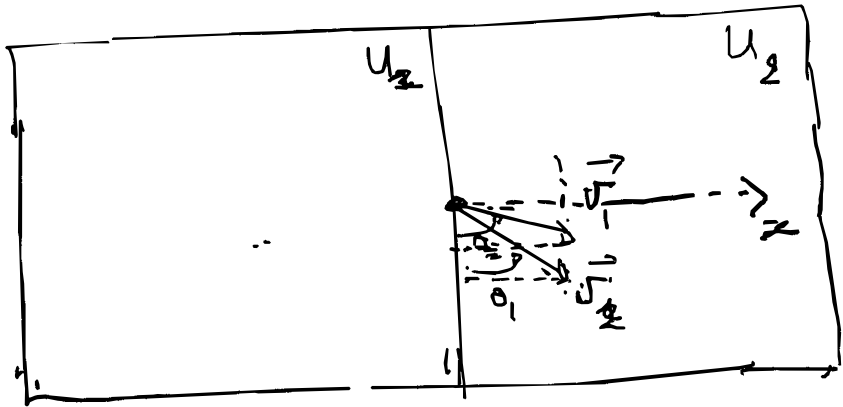
Remark:

$$L = \frac{1}{2} \sum_a m_a v_a^2 - U(\{r_a\}_{a \in A})$$

$$r_{a'} \rightarrow r_{a'} + \varepsilon \Rightarrow \delta L = 0$$

if the potential energy does not depend on a coordinate direction then momentum is conserved in that direction!

Problem 4 :



Conservation of energy:

$$E = \frac{1}{2} m \|\vec{v}_1\|^2 + U_1 = \frac{1}{2} m \|\vec{v}_2\|^2 + U_2$$

$$\frac{1}{2} m \|\vec{v}_1\|^2 + U_1 - U_2 = \frac{1}{2} m \|\vec{v}_2\|^2$$

Conservation of Momentum $\vec{e}_x$

$$v_{1x} = v_{2x} \Rightarrow v_1 \sin \theta_1 = v_2 \sin \theta_2$$

$$\left(\frac{\sin \theta_1}{\sin \theta_2} \right)^2 = \frac{v_2^2}{v_1^2} = \frac{v_1^2 + \frac{2}{m} (U_1 - U_2)}{v_1^2}$$

$$\Rightarrow \left| \frac{\sin \theta_1}{\sin \theta_2} \right| = \sqrt{1 + \frac{2}{m v_1^2} (U_1 - U_2)}$$